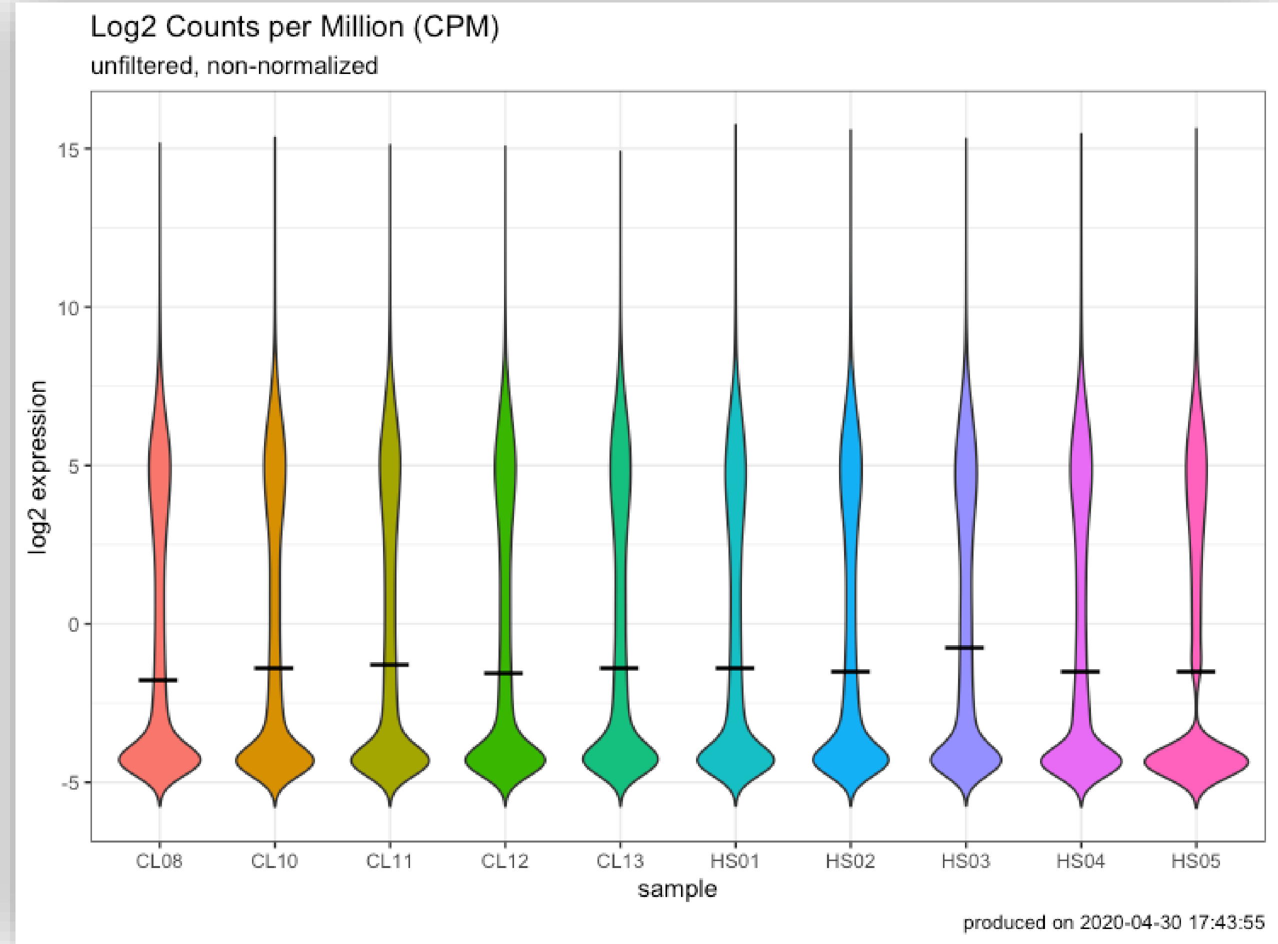
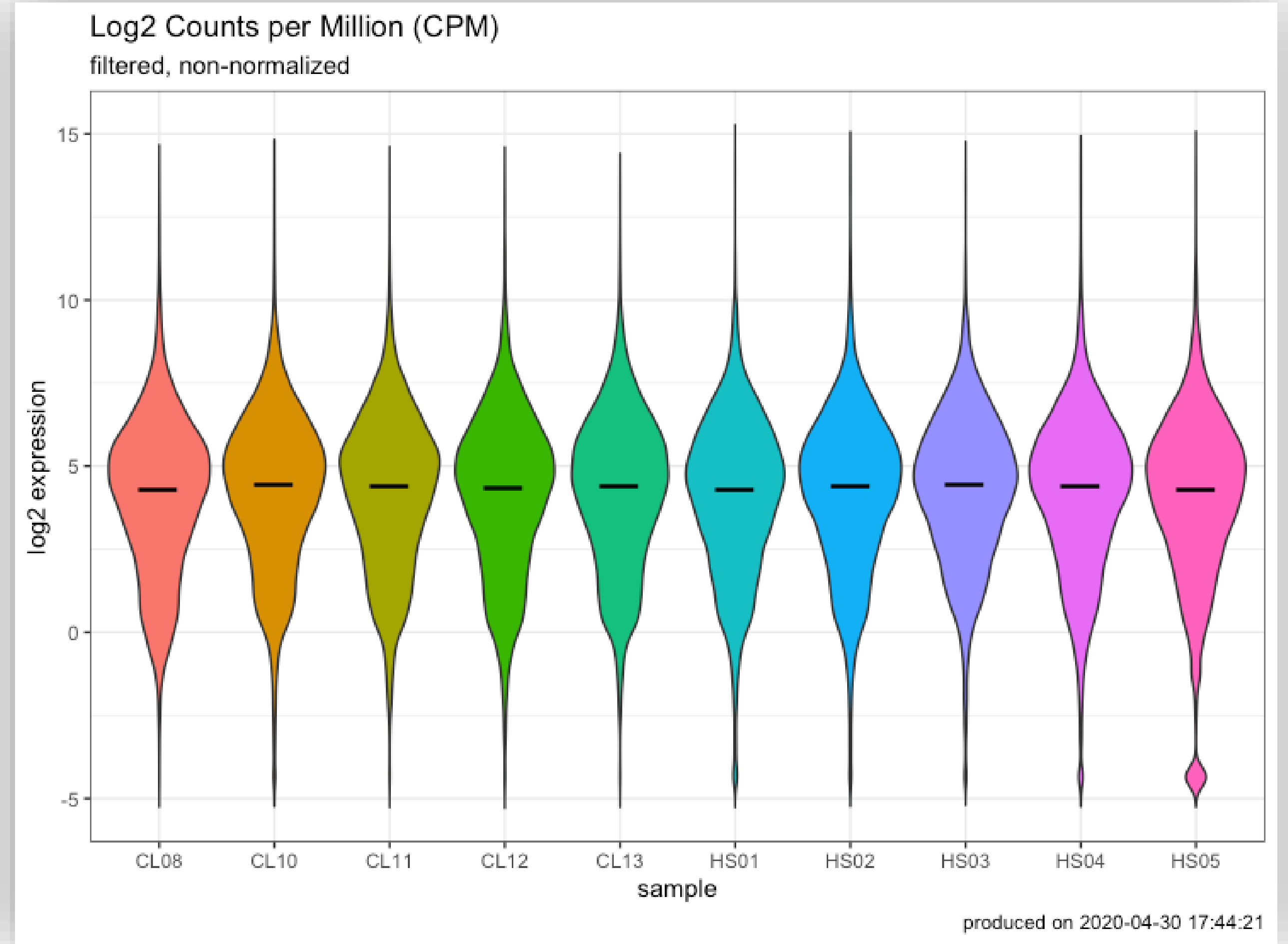
Lecture 9: Differential Gene Expression

You've been preparing for DEG analysis all along

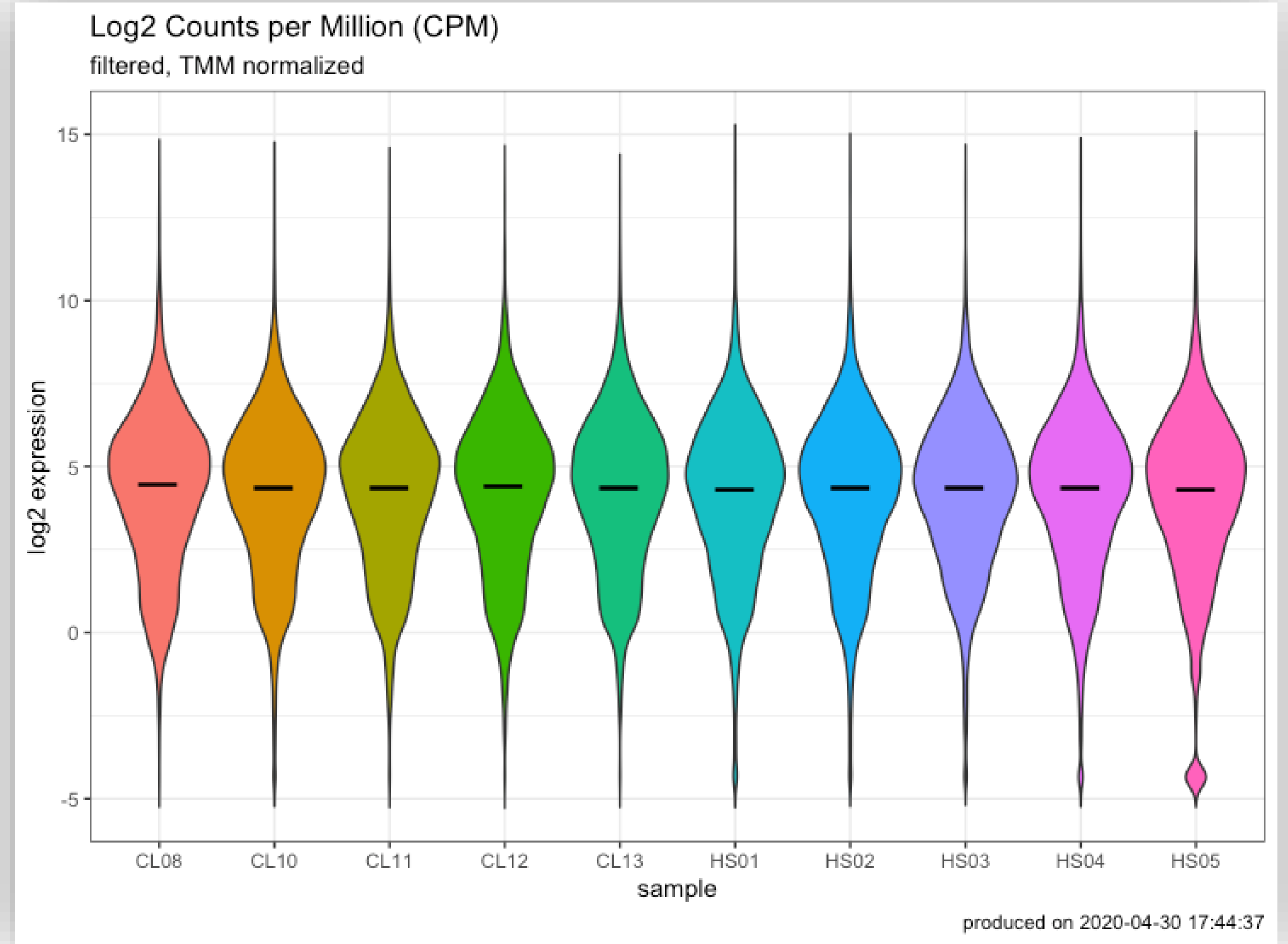
raw



filtered

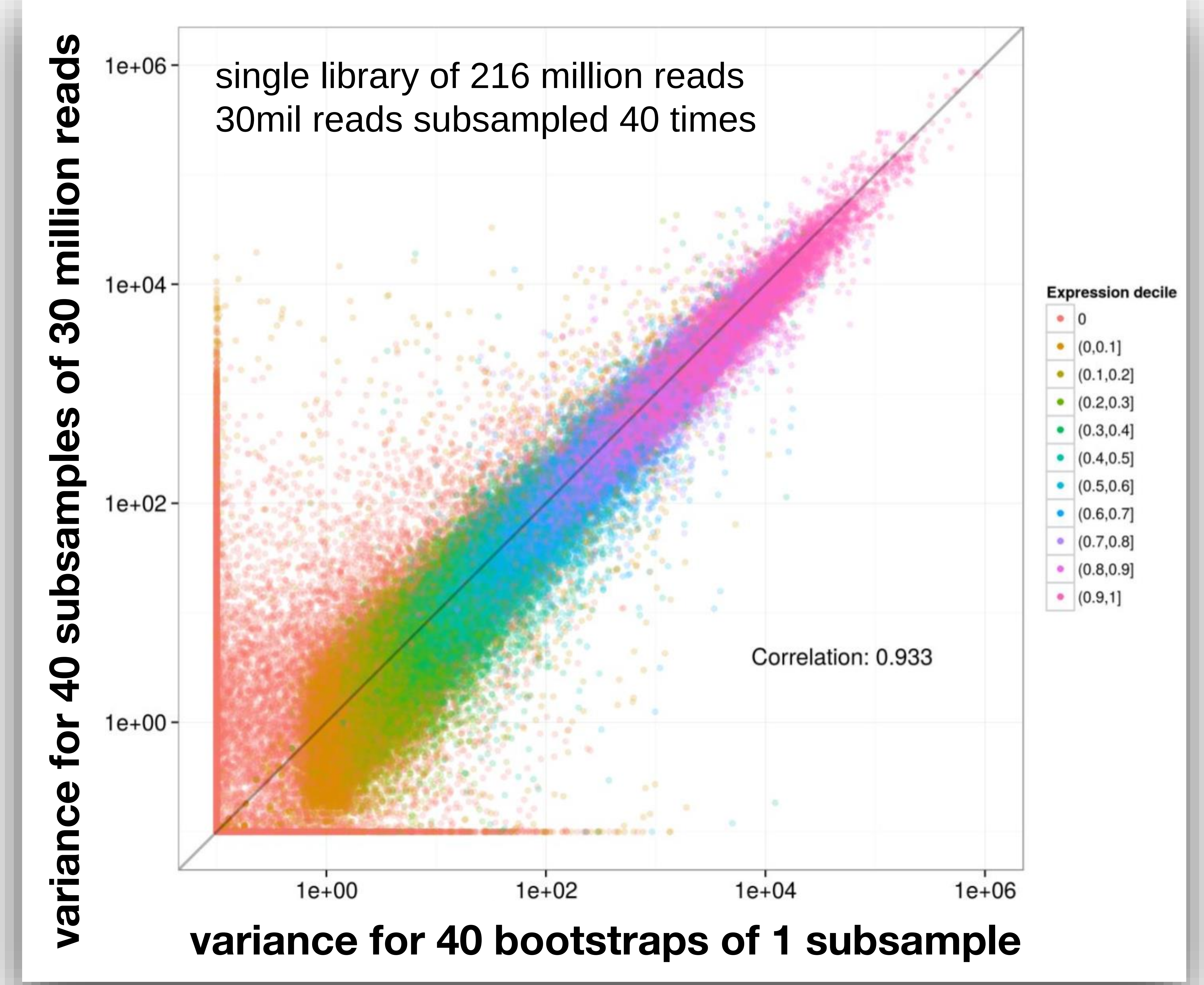


normalized



Uncertainty in RNAseq counts

Kallisto bootstraps (-b)



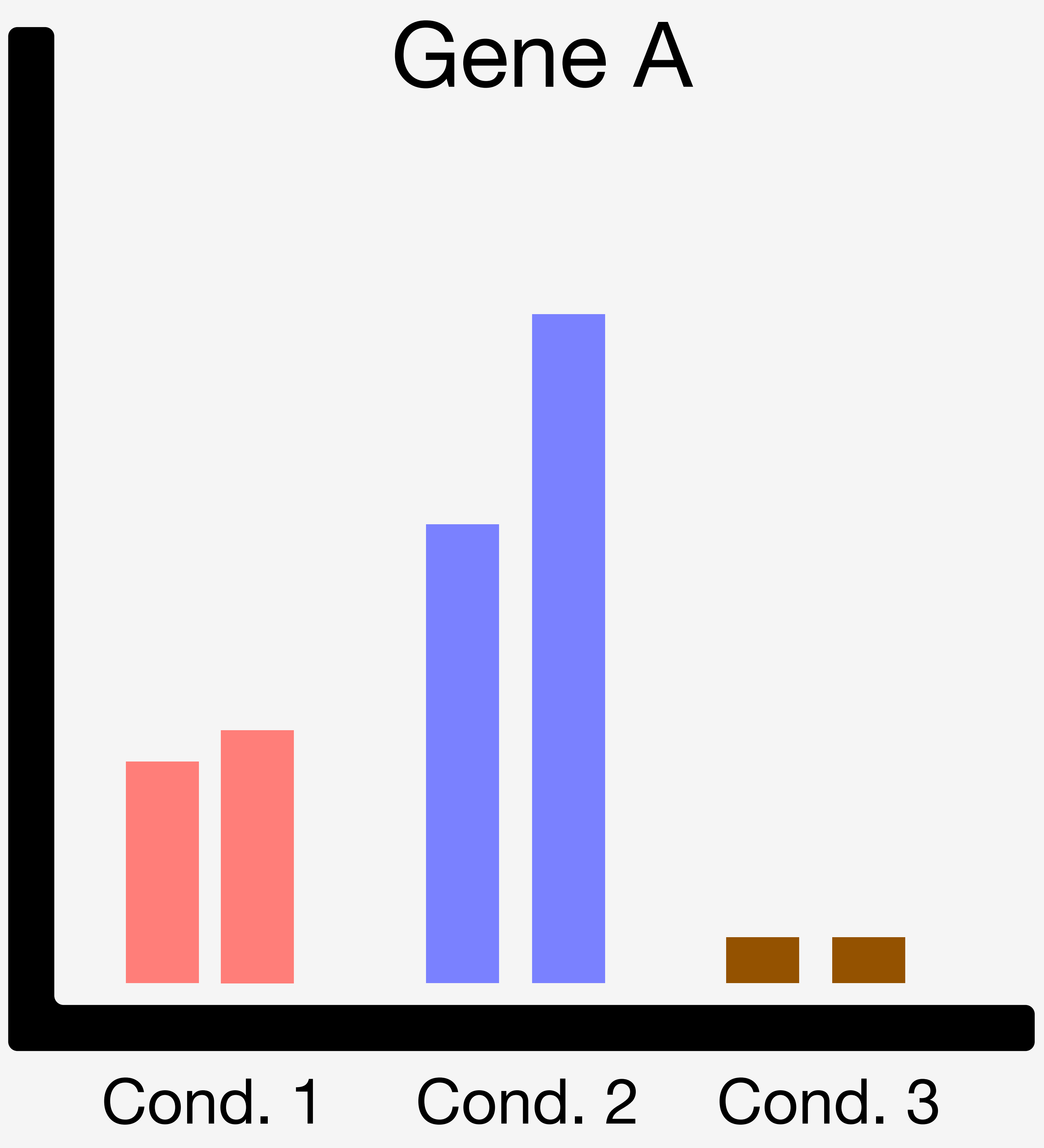
Bray et al., Nat. Biotech., 2016

Some terminology

DGE
Differential Gene Expression

DTE
Differential Transcript Expression

DTU
Differential Transcript Usage

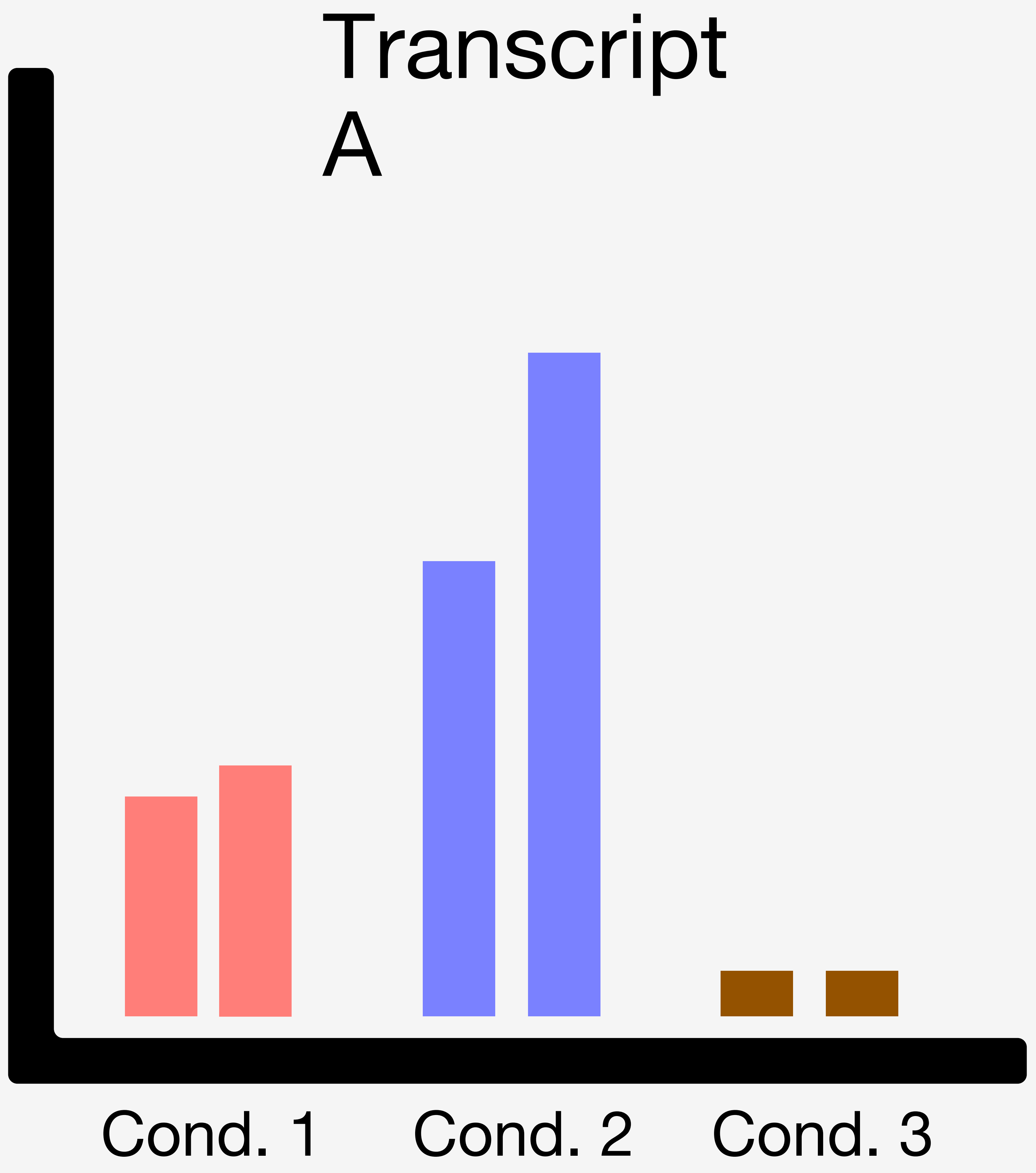


Some terminology

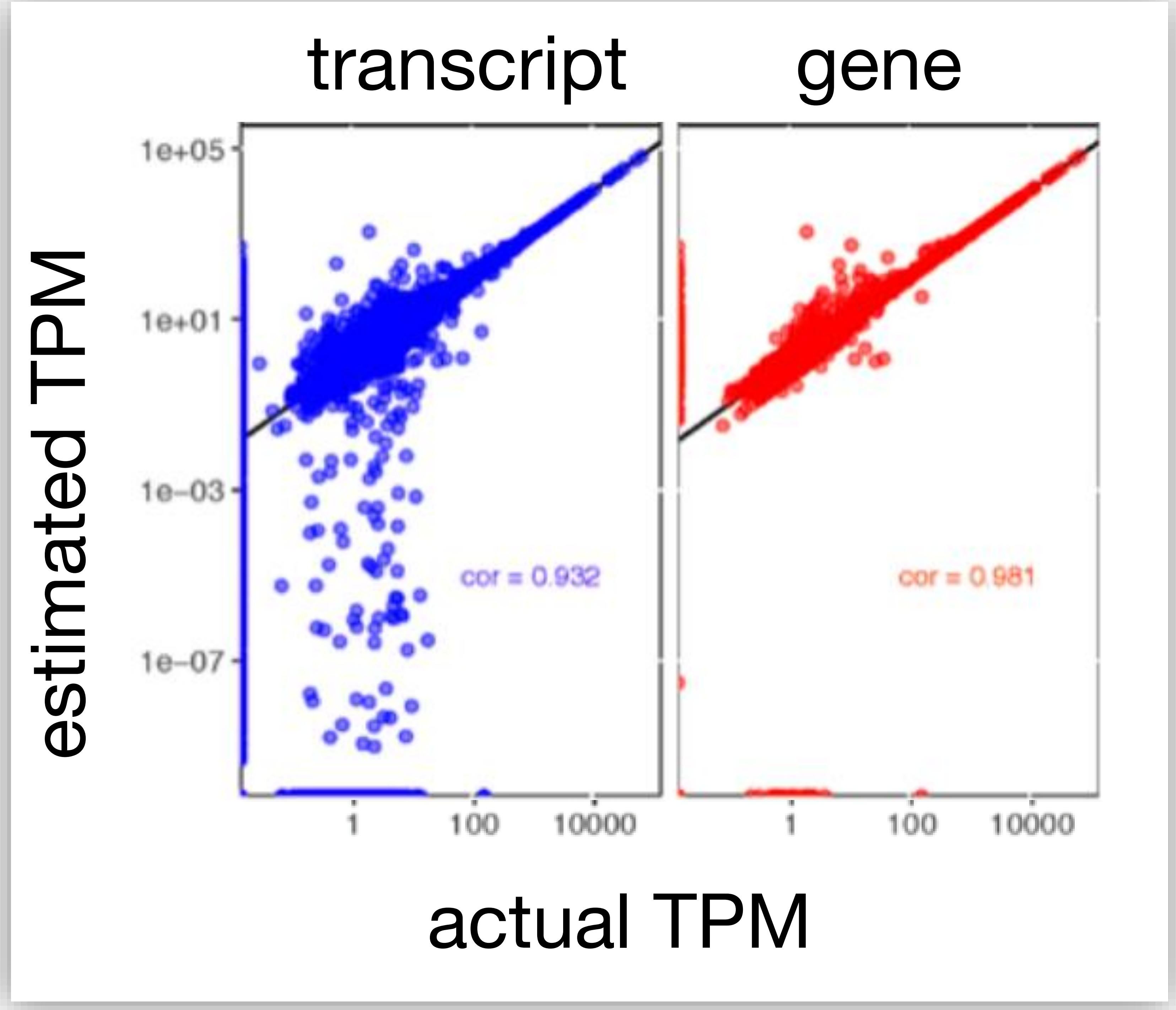
DGE
Differential Gene Expression

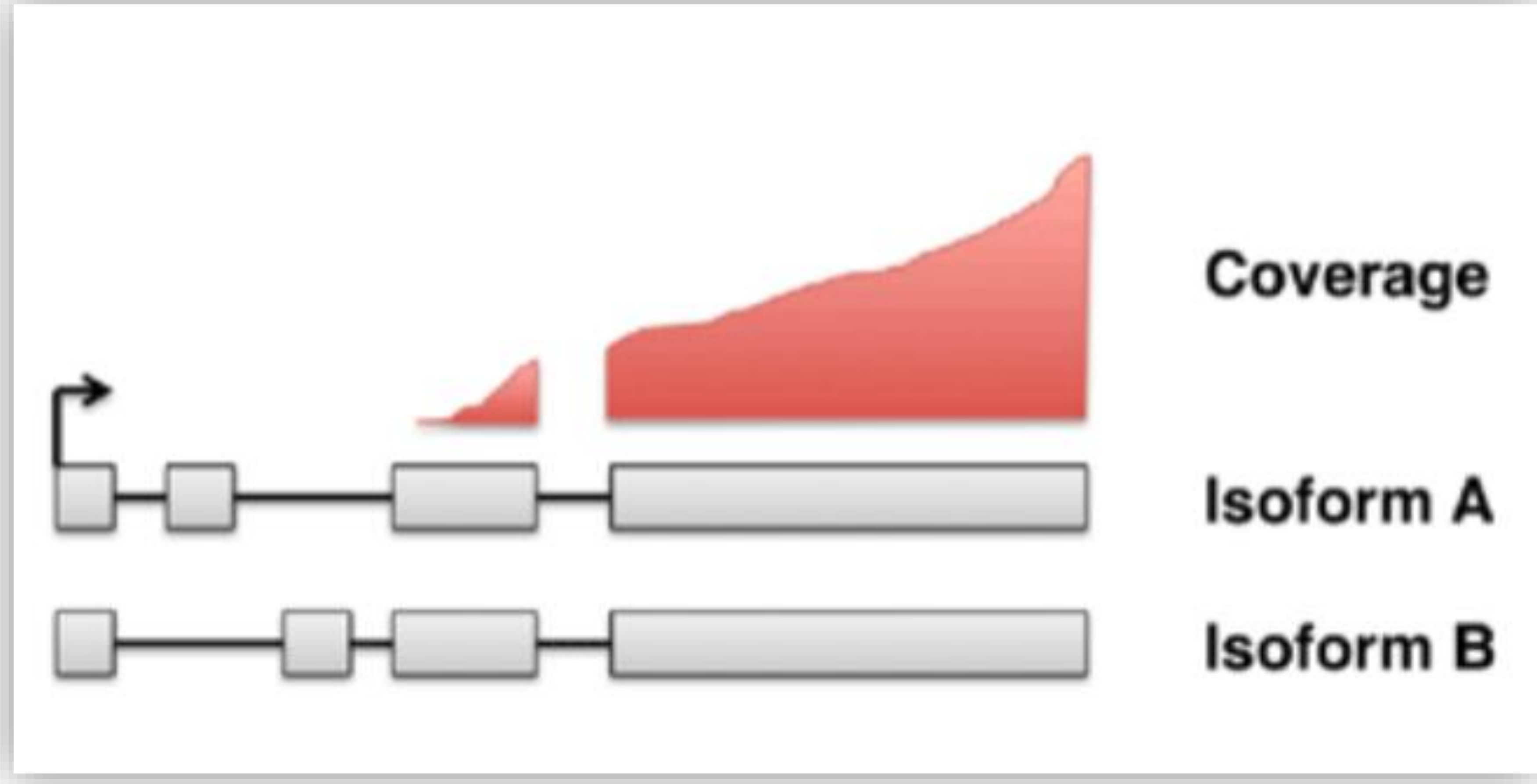
DTE
Differential Transcript Expression

DTU
Differential Transcript Usage



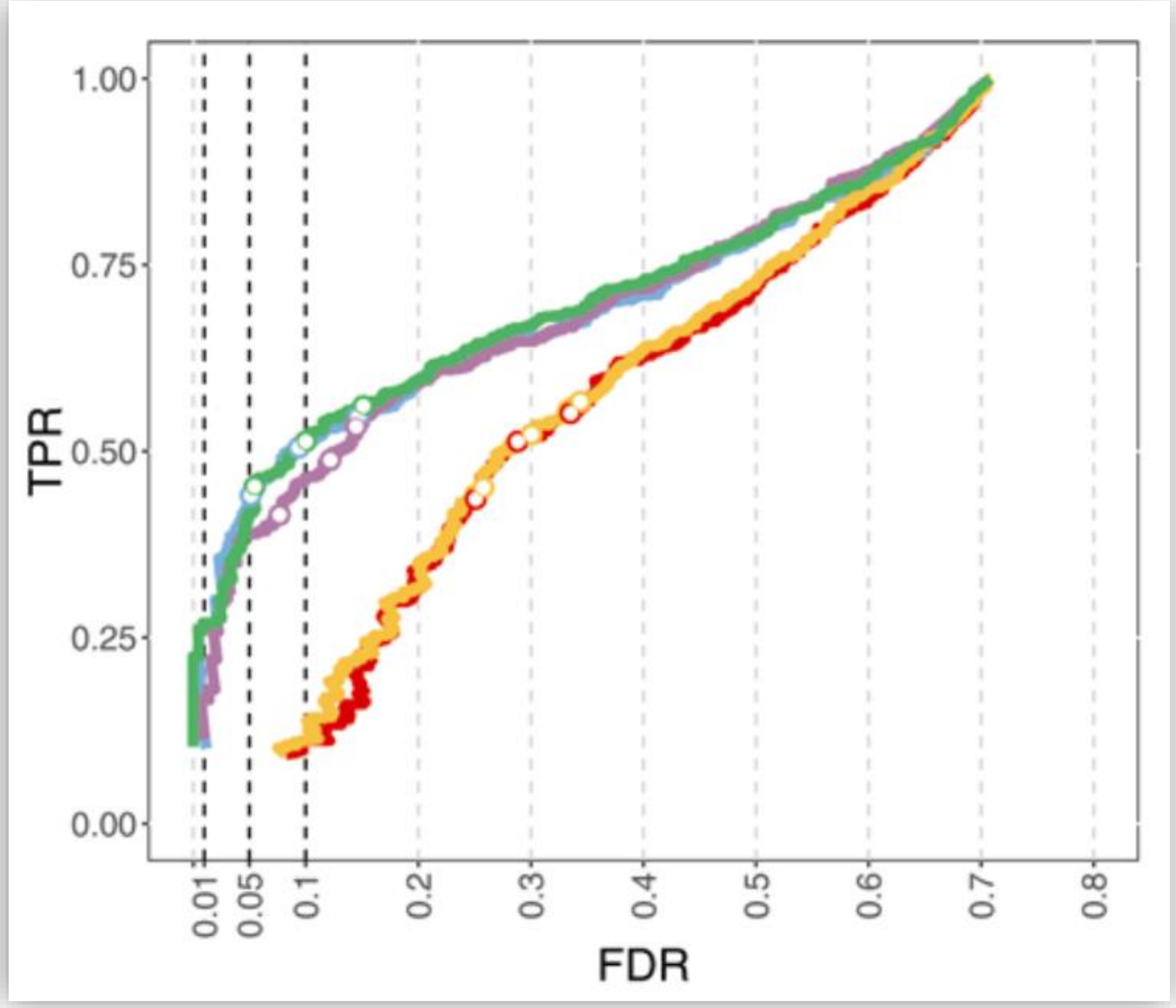
Genes can be more accurately measured than transcripts





Soneson et al., F1000Research., 2016

Aggregate transcript data to the gene level



Soneson et al., F1000Research., 2016

Some terminology

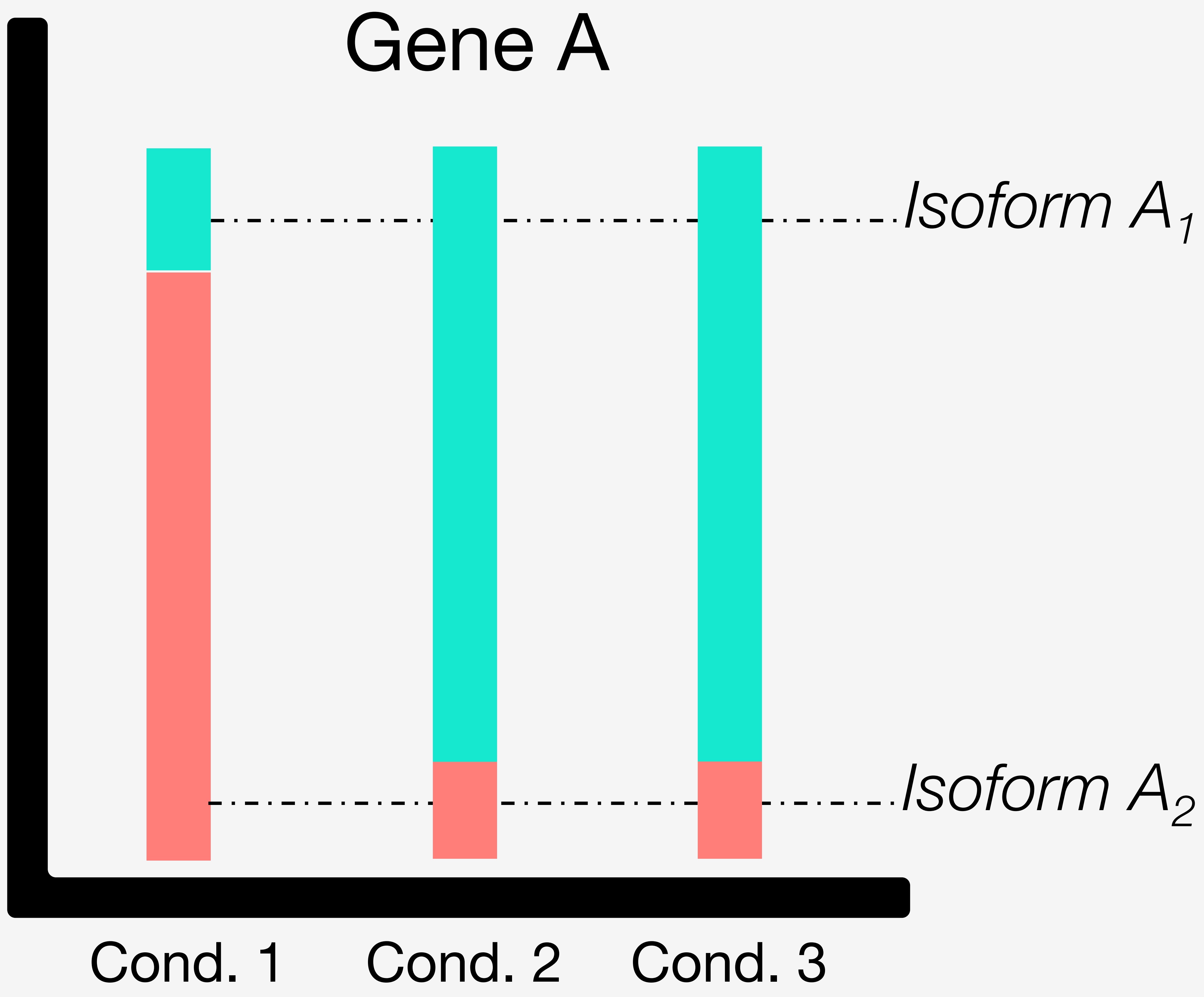
DGE
Differential Gene Expression

DTE
Differential Transcript Expression

Not mutually exclusive

DTU
Differential Transcript Usage

"Isoform switching"



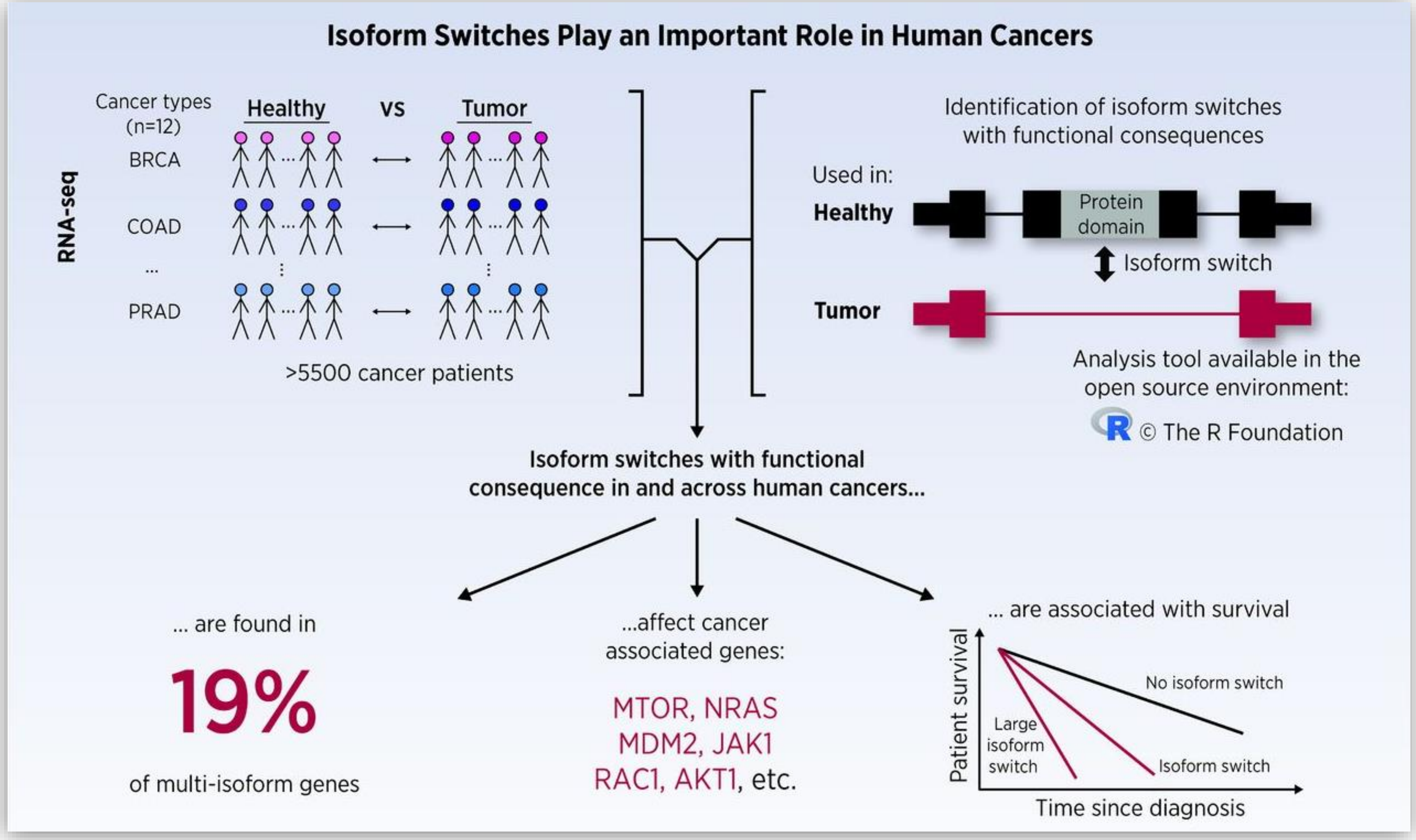
Choosing your path

Task	Input data	Software (examples)	Post-processing
DGE	Aggregated transcript counts + average transcript length offsets, or simple counts + average transcript length offsets	Salmon, kallisto, BitSeq, RSEM	
		tximport	
		DESeq2, edgeR, voom/limma	
DTE	Transcript counts	Salmon, kallisto, BitSeq, RSEM	Optional gene-level aggregation
		tximport	
		DESeq2, edgeR, sleuth, voom/limma	
DTU/DEU	Transcript counts or bin counts, depending on interpretation potential[18]	Salmon, kallisto, BitSeq, RSEM	Optional gene-level aggregation
		DEXSeq	

Soneson et al., F1000Research., 2016

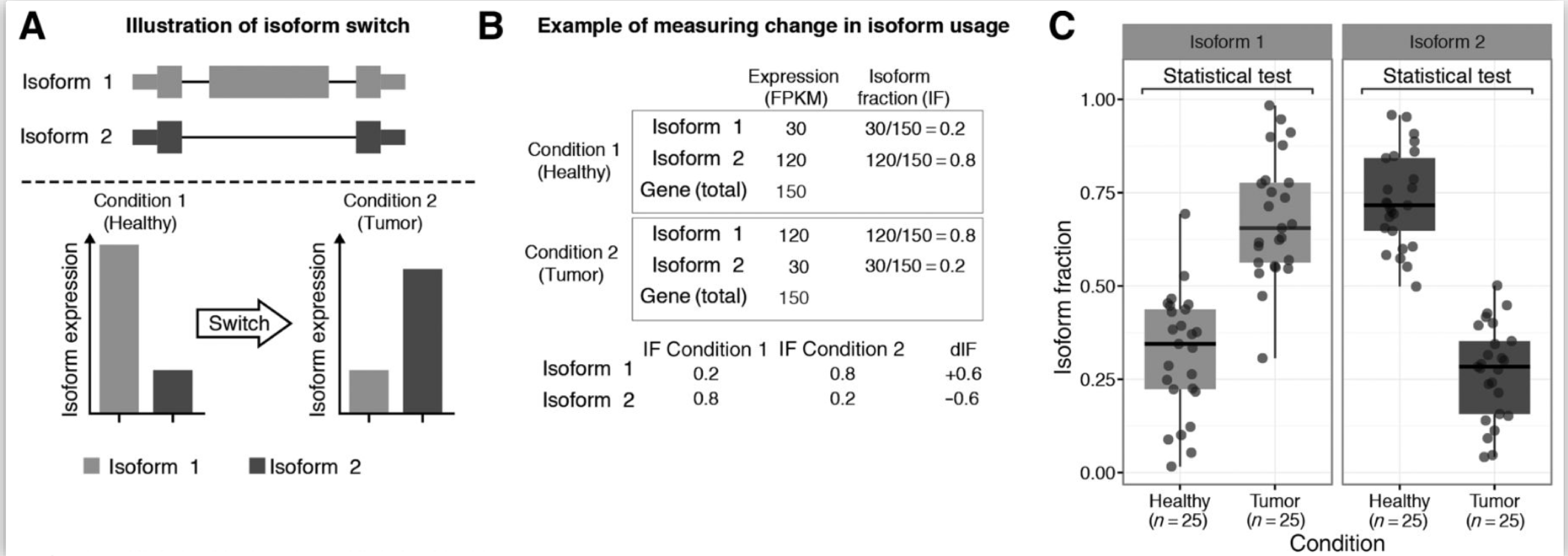
Start step 4 script

Isoform switching and cancer



Vitting-Seerup and Sandelin, Molecular Cancer Research. 2016

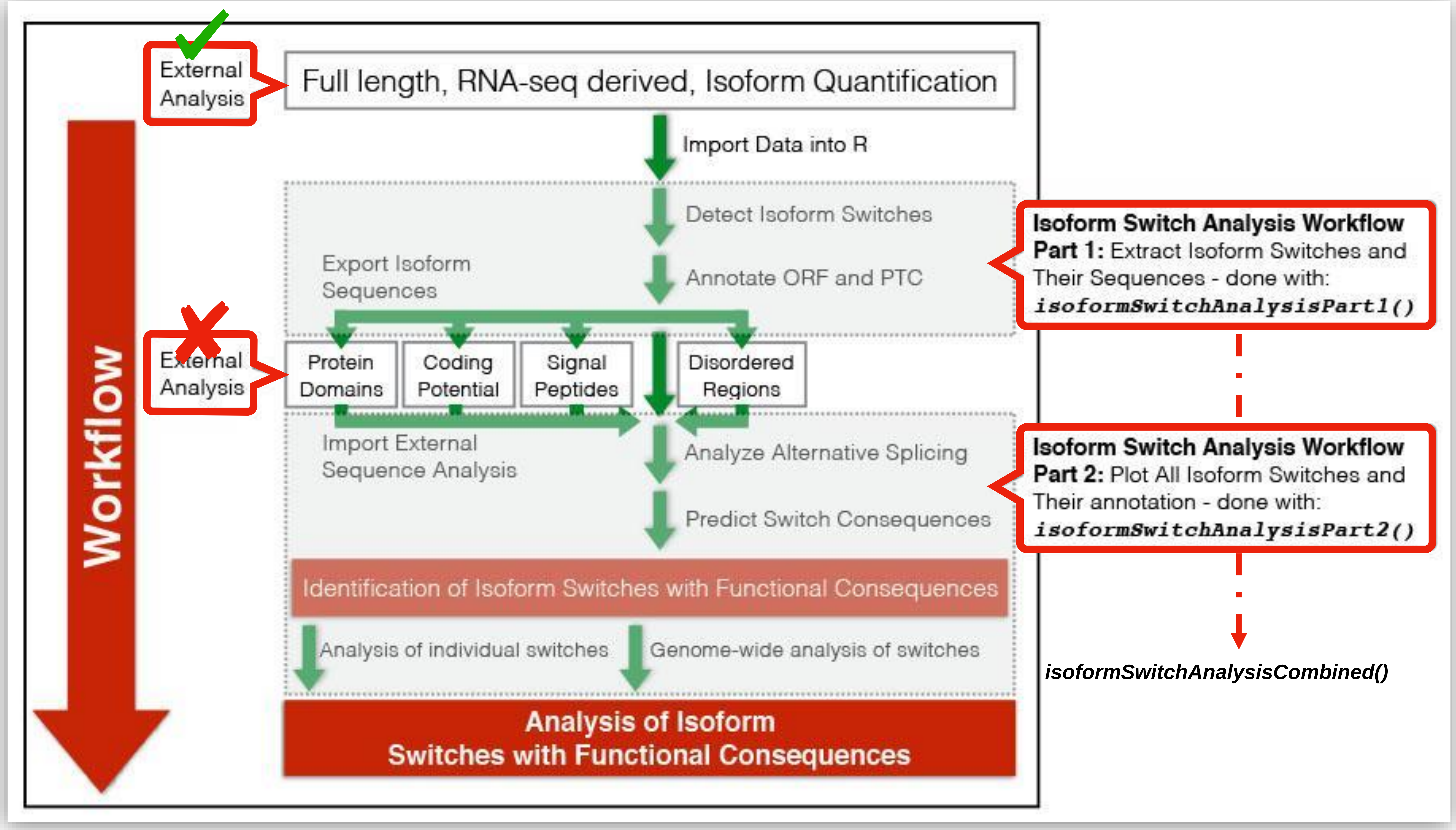
Defining and detecting an isoform switch



Vitting-Seerup and Sandelin, Molecular Cancer Research. 2016

IsoformSwitchAnalyzeR

high-level functions for DTU analysis



Vitting-Seerup and Sandelin, Bioinformatics. 2019

Warning!

DTU analysis with *isoformSwitchAnalyzeR* will not work unless you upgrade R <u>and</u> bioconductor

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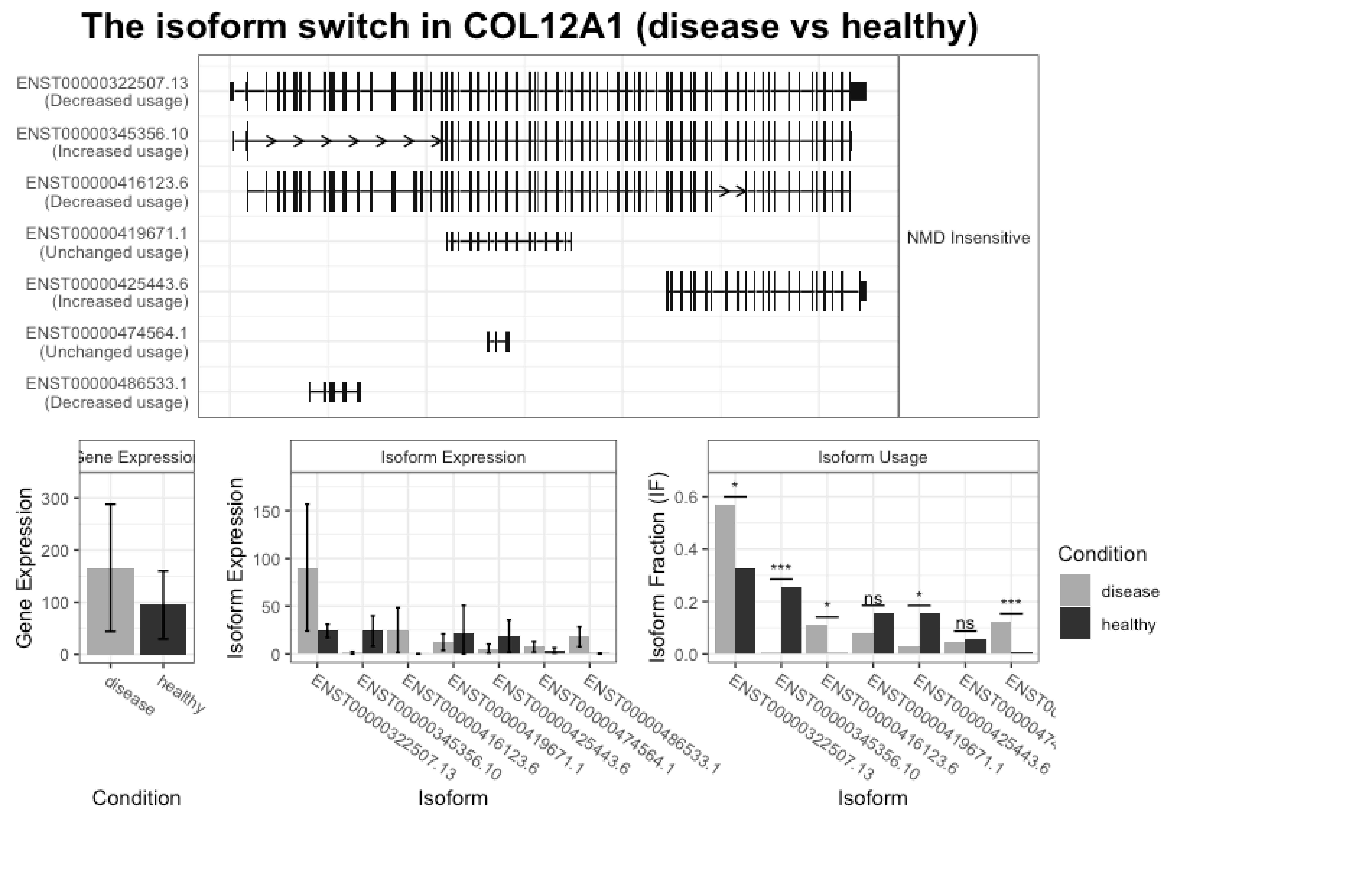
GTF files provide a map for all known genome features

Column	Description
Seqname	Name of chromosome, scaffold or contig
Source	Name of the datasource
Feature	Feature type (e.g. gene, exon, transcript, CDS, five_prime_UTR)
Start	Start position of the feature
End	End position of the feature
Empty	‘.’
Strand	Defined as + (forward) or - (reverse)
Frame	Either ‘0’, ‘1’, or ‘2’ - indicating reading frame by base position
Attribute	A semicolon separated list of other info about the feature

DTU example


The isoform switch in COL12A1 (disease vs healthy)
ENST00000322507.13
(Decreased usage)
ENST00000345356.10
(Increased usage)
ENST00000416123.6
(Decreased usage)
ENST00000419671.1
(Unchanged usage)
ENST00000425443.6
(Increased usage)
ENST00000474564.1
(Unchanged usage)
ENST00000486533.1
(Decreased usage)
NMD Insensitive
Gene Expression
300
200
100
0
disease
healthy
Condition
Isoform Expression
150
100
50
0
ENST00000322507.13
ENST00000345356.10
ENST00000416123.6
ENST00000419671.1
ENST00000425443.6
ENST00000474564.1
ENST00000486533.1
Isoform
Isoform Usage
Isoform Fraction (IF)
0.6
0.4
0.2
0.0
*
***
*
ns
*
ns
***
Condition
disease
healthy

DGE + DTE + DTU example